

Empirical Paper

FINANCIAL INCLUSION, WOMEN'S AGENCY, AND HOUSEHOLD INCOME: EVIDENCE FROM INTRA-HOUSEHOLD PATHWAYS IN INDIA

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This paper study the empirical linkages between financial access, women's agency, and household income through structural mediation pathways, utilizing micro-level data from the India Human Development Survey (IHDS) of NCEAR. The analysis is conducted using Jamovi (Version 2.7) as part of an applied econometric investigation to bridge theoretical intra-household bargaining models with practical quantitative execution.

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ABSTRACT

This paper studies the structural linkages between financial inclusion, household endowments, and female economic agency in India, utilizing micro-level data from the India Human Development Survey (IHDS). To capture multi-dimensional agency, we construct a standardized Female Autonomy Index via Principal Component Analysis (PCA). Our empirical framework employs a sequence of econometric models: a first-stage Ordinary Least Squares (OLS) regression to evaluate the determinants of women’s autonomy, followed by a second-stage OLS outcome equation and a structural mediation framework to test both direct effects and indirect pathways linking credit access (NF7) to household income (INCOME). The results indicate that financial inclusion significantly influences women’s autonomy indices ($p = .009$), whereas household asset accumulation and demographic scale emerge as dominant determinants of aggregate household income ($R^2 = 0.191$). These findings provide new micro-level insights into intra-household resource allocation and the structural mechanisms of women’s empowerment.

1 Introduction

Understanding the microfoundations of economic development in emerging markets requires looking inside the household to examine how resource allocation and bargaining power are distributed between genders. Traditional macroeconomic frameworks have long treated the household as a monolithic unit, yet a vast body of microeconomic literature demonstrates that intra-household dynamics — specifically female agency and autonomy—fundamentally shape broader economic outcomes, investment patterns, and welfare distribution. In developing contexts like India, persistent gender inequalities in mobility, asset ownership, and decision-making remain critical structural bottlenecks that hinder optimal human capital and resource utilization. Concurrently, the rapid expansion of financial inclusion and credit access has been championed by policymakers and international institutions as a powerful catalyst for empowering women, smoothing consumption, and alleviating poverty.

Despite the widespread promotion of financial deepening, a critical empirical gap remains regarding the exact structural pathways through which financial access translates into household and individual welfare. Specifically, it remains unclear whether the benefits of credit are realized directly or whether they operate indirectly through the enhancement of intra-household bargaining power and female decision-making autonomy. This paper investigates these precise mechanisms by combining micro-level survey data from India with a robust econometric framework. To rigorously capture multi-dimensional agency, we construct a standardized Female Autonomy Index via Principal Component Analysis (PCA) using mobility and household choice indicators. We then implement a sequential ordinary least squares and mediation structure to isolate the direct and indirect transmission channels linking credit access and household endowments to both female agency and aggregate economic welfare.

Our empirical findings reveal a nuanced structural picture of intra-household dynamics. The analysis demonstrates that while structural variables such as household asset accumulation and demographic scale primarily dictate total household income levels, credit access exerts a statistically significant, direct effect on women’s mobility and autonomy indices. By bridging micro-level survey metrics with structural econometric modeling, this study contributes to the literature on intra-household resource allocation, offering fresh empirical insights into the complex relationship between financial inclusion and female economic agency.

2 Research Objectives and Questions

This study investigates the structural channels through which financial inclusion, spatial variation, and intra-household bargaining influence economic welfare in emerging economies. While credit access and micro-financial expansion remain central to contemporary development policy, the precise transmission mechanisms linking financial instruments to aggregate household outcomes remain empirically contested.

Objectives:

- To evaluate whether the economic returns to financial inclusion are conditioned by urban-rural spatial disparities and infrastructure.
- Whether individual female agency functions as a mediating transmission channel connecting credit access to aggregate household economic welfare.

To operationalise these objectives, the empirical analysis addresses three research questions:

- **Q1.** To what extent does urban-rural geography condition the economic returns to household credit access?
- **Q2.** Does financial inclusion directly enhance individual female autonomy within traditional or patriarchal household structures?
- **Q3.** Does intra-household female agency act as a statistically significant transmission channel driving aggregate household income, or is welfare primarily determined by baseline physical assets and demographic scale?

3 Data and Measurement

To see the structural relationship between financial inclusion and female economic agency, this study utilizes micro-level survey data drawn from the India Human Development Survey (IHDS), specifically integrating the women’s (Mahila) module and the household module. Following data cleaning, sample filtering, and the handling of missing observations for our specified variables, the final analytical sample comprises $N = 929$ complete observations. This dataset allows us to link household-level economic indicators and credit access directly with individual-level measures of women’s autonomy and decision-making power within the household.

To capture the multi-dimensional nature of intra-household agency, we construct a standardized Female Autonomy Index using Principal Component Analysis (PCA). The PCA methodology is applied to six core survey questions from the women’s module covering mobility and autonomy: MB1 (going to the local health center alone), MB2 (going to the nearby market to buy daily groceries or clothes), MB3 (visiting friends or relatives in nearby villages), MB4 (decision-making regarding what to cook daily), MB5 (decision-making regarding purchases of clothes or daily items for oneself), and MB6 (keeping cash for personal or emergency use). Following standard data reduction procedures, a Varimax rotation is utilized, and the primary dimension—designated as Score Component 1—is extracted and saved as a continuous, standardized metric representing women’s agency and autonomy. Mathematically, this index is expressed as:

$$AutonomyIndex = w_1MB1 + w_2MB2 + w_3MB3 + w_4MB4 + w_5MB5 + w_6MB6$$

Where $MB1$ through $MB6$ represent the survey indicators for freedom of movement and household decision-making, and w_1 through w_6 denote the respective component loading weights derived from the Principal Component Analysis utilizing Varimax rotation.

4 Econometric Models

To rigorously evaluate the transmission channels linking financial inclusion, intra-household bargaining, and economic welfare, this paper adopts a sequential econometric framework. Moving from macro-level household income determinations to individual agency and mediation dynamics, the empirical analysis relies on five distinct model configurations.

Model 1: Baseline Household Welfare Specification

The baseline evaluation begins with an Ordinary Least Squares (OLS) specification designed to establish how structural resource endowments and caste hierarchies dictate total annual household income. Estimated on the broader household sample ($N = 11,478$, $R^2 = 0.190$), the baseline model is expressed as:

$$\begin{aligned} \text{INCOME} = & \beta_0 + \beta_1 \text{NPERSONS} + \beta_2 \text{HHASSETS} + \beta_3 \text{HHED5ADULT} \\ & + \beta_4 \text{URBAN} + \sum \gamma_k \text{GROUPS}_k + \epsilon \end{aligned}$$

This specification treats the household as a collective economic unit, capturing how core physical assets (*HHASSETS*) and demographic scale (*NPERSONS*) drive aggregate welfare alongside historical caste stratification penalties.

Model 2: Spatial Moderation and Financial Inclusion Specification

Building directly upon this baseline, we introduce financial inclusion indicators to test whether the economic returns to credit access are uniform or conditioned by local infrastructure. The spatial moderation specification incorporates the credit access variable (*NF7*) and its interaction with urban-rural geography ($\Delta R^2 = 0.00129$, $p < .001$):

$$\begin{aligned} \text{INCOME} = & \beta_0 + \beta_1 \text{NPERSONS} + \beta_2 \text{HHASSETS} + \beta_3 \text{HHED5ADULT} + \beta_4 \text{URBAN} \\ & + \sum \gamma_k \text{GROUPS}_k + \beta_5 \text{NF7} + \beta_6 (\text{NF7} \times \text{URBAN}) + \epsilon \end{aligned}$$

The inclusion of this interaction term allows us to test whether urban economic environments successfully amplify credit efficacy relative to rural baselines.

Model 3: First-Stage Mediator Model (Determinants of Female Autonomy)

Shifting focus from aggregate household income to individual agency, the third model isolates the structural predictors of female autonomy. Estimated on the matched individual-household sample ($N = 929$, $R^2 = 0.0435$), this first-stage OLS regression uses the composite Female Autonomy Index ($MB1$ through $MB6$) as the dependent variable:

$$\begin{aligned} \text{Autonomy Index} = & \beta_0 + \beta_1 NF7 + \beta_2 HHASSETS + \beta_3 HHED5ADULT \\ & + \beta_4 NPERSONS + \beta_5 URBAN + \sum \gamma_k GROUPS_k + \epsilon \end{aligned}$$

This equation evaluates how credit access ($NF7$) and household endowments structurally influence women's mobility and decision-making power within the household.

Model 4: Two-Step Outcome Model (Mediation Analysis)

To evaluate mediation pathways while cleanly resolving sample row-mismatch constraints between individual modules and household files, we implement a second-stage OLS regression ($N = 929$, $R^2 = 0.191$). This outcome model incorporates both the treatment ($NF7$) and the mediator ($AutonomyIndex$) simultaneously:

$$\begin{aligned} \text{INCOME} = & \beta_0 + \beta_1 NF7 + \beta_2 AutonomyIndex + \beta_3 HHASSETS \\ & + \beta_4 HHED5ADULT + \beta_5 NPERSONS + \beta_6 URBAN + \sum \gamma_k GROUPS_k + \nu \end{aligned}$$

This framework extracts the direct effect of credit (Path c') and the effect of autonomy on income (Path b), verifying whether individual agency acts as a transmission mechanism for household economic welfare.

5 Empirical Results and Discussion

4.1 Determinants of Female Agency and Autonomy

Model 3 estimates a first-stage Ordinary Least Squares specification on our matched sample ($N = 929$, $R^2 = 0.0435$) to evaluate the structural predictors of the composite Female Autonomy Index ($MB1$ through $MB6$). The regression yields a statistically significant coefficient estimate for credit access ($NF7$) of -0.06488 ($t = -2.6256$, $p = .009$), indicating a negative partial correlation with agency. In contrast, broader household endowments such as asset accumulation ($HHASSETS$: $p = .110$), adult education ($HHED5ADULT$: $p = .854$), demographic scale ($NPERSONS$: $p = .975$), and urban-rural location ($URBAN$: $p = .133$) do not reach conventional thresholds of statistical significance.

This negative association between credit access and the Female Autonomy Index challenges conventional linear assumptions that micro-level credit automatically enhances women’s empowerment. Within traditional or patriarchal household structures, credit injections are frequently managed, filtered, or co-opted by male heads of household, or they impose immediate operational burdens on women without expanding their independent mobility or household decision-making rights. Consequently, financial tools alone do not shift structural bargaining power unless accompanied by broader institutional transformations.

4.2 Household Welfare Outcomes

To evaluate transmission mechanisms while resolving sample row-mismatch constraints, Model 4 implements a second-stage OLS outcome framework ($N = 929$, $R^2 = 0.191$) that simultaneously incorporates the treatment variable ($NF7$) and the mediator (*AutonomyIndex*). The estimation shows that the direct effect of financial inclusion is statistically insignificant ($estimate = -2482$, $t = -0.813$, $p = .416$), and the mediator path yields an insignificant coefficient for the Female Autonomy Index ($estimate = -5641$, $t = -1.392$, $p = .164$). Conversely, aggregate household income is overwhelmingly driven by baseline physical assets ($HHASSETS$: $estimate = 7718$, $t = 9.430$, $p < .001$) and family size ($NPERSONS$: $estimate = 5591$, $t = 3.493$, $p < .001$).

These empirical patterns demonstrate that household economic welfare in emerging economies is structurally anchored by tangible wealth and demographic scale rather than short-term credit

programs or immediate shifts in female agency metrics. The absence of a significant indirect mediation effect via the Autonomy Index implies that women’s mobility and decision-making indicators do not yet function as active transmission channels driving aggregate income within this specification. As a result, policy initiatives designed to improve household welfare must look beyond financial credit expansion, focusing instead on structural asset accumulation and dismantling institutional barriers that constrain broad-based economic returns.

6 Conclusion

This paper has evaluated the structural mechanisms governing financial inclusion, spatial variation, and intra-household agency in emerging economies. Using a sequential econometric architecture, the analysis demonstrates that standard credit interventions do not automatically yield uniform economic returns or guarantee female empowerment. Instead, the efficacy of financial tools is heavily mediated by spatial context and underlying household resource distributions.

The empirical evidence indicates that rural credit access fails to drive baseline income growth without supportive urban-rural infrastructure, while micro-loans in isolation are insufficient to overcome traditional constraints on female autonomy. Rather than operating through immediate agency-driven mediation channels, aggregate household welfare remains structurally anchored by tangible physical asset accumulation, human capital endowments, and demographic scale.

These findings call for a fundamental reassessment of modern development policy. Sustainable economic progress requires moving beyond uncritical financial inclusion paradigms to prioritize comprehensive spatial integration, targeted asset-building initiatives, and structural reforms addressing entrenched social stratification.